

Math 221D Fall 2026

General information

- Instructor: Danielle Wang, danielle.wang@duke.edu
- Discussion section leader: Alina Zhu
- Danielle's office hours: Monday and Wednesday, 9:50am–11am, Physics 274E
- Alina's office hours: Thursday, 5pm–7pm, Gross Hall 304B
- Math helproom: Monday, 7pm–9pm, Classroom Building 136; Tuesday–Thursday, 7pm–9pm, Physics 274
- Required textbook: *Linear Algebra: A Geometric Approach* (2nd Edition) by Theodore Shifrin and Malcolm Adams
- Lectures: Monday and Wednesday, 8:30am–9:45am, Physics 259
- Discussions: Friday, 8:30am–9:45am, Physics 047
- Website: <https://danielleywang.github.io/221dfa26>

Exams

Exams will make up 80% of your grade in the class. There will be **two in-class midterms**, held during discussion section. Midterm 1 will be worth 20% and Midterm 2 will be worth 25%. There will also be a **final exam**, worth 35%. The dates for the midterm exams and the final are below.

- Midterm 1: Fri Oct 2
- Midterm 2: Fri Nov 6
- Final Exam: Sun Dec 13, 9am–12pm, Physics 259

Quizzes

There will also be weekly quizzes given during discussion section, worth 10% of your grade in the class. Your lowest quiz score will be dropped.

All exams and quizzes will be closed book.

Homework

Homework will make up the remaining 10% of your grade in the class. Your lowest two homework scores will be dropped.

Problem sets will be posted and submitted on Gradescope and will (tentatively) be due on **Thursdays** at 11:59pm. The first pset is due on September 2, 2026.

Solutions to homework and exam problems should be logically complete and clearly written in order to receive credit. When appropriate, complete sentences should be used to develop arguments.

Late homework

Homework may be submitted up to **two days** past the due date; late submissions will receive a 10% penalty.

Collaboration and outside resource policy

Collaboration with classmates on problem sets is allowed, but your solutions must be written by yourself.

At the top of your submission for each problem set, please indicate all resources other than the course textbook that provided substantial assistance with your solutions.

For example, you should list the names of students with whom you discussed the problems, any large language models you used, and websites such as math.stackexchange.com or Wikipedia.

AI policy

- Use of LLM's for the homework is permitted, although not encouraged. If you do use LLM's, we suggest that you use them to interactively to help you work towards solutions to homework problems (rather than simply copying the entire problem statement and asking for an answer).
- Remember that (as a special case of the collaboration policy) the writing of the solution must be in your own words; it is unacceptable to copy-paste the output of an LLM and submit it.

Grading

Final letter grades will be based, at the instructor's discretion, on students' performance across exams, quizzes, and homework. The cutoffs for an A-, B-, and C- will be no higher than 90%, 80%, and 70%, respectively, but may be adjusted.

Missed work

Homework extensions will not be granted (beyond the two-day grace period), and missed exams and quizzes cannot be made up, except in the cases below.

Excused absences

- If a student is too ill to complete an assignment or attend a test, they should inform the instructors as soon as possible using the on-line [Short Term Incapacitation Form](#).
- Students who miss tests or assignments due to a scheduled varsity athletic trip or religious holiday should submit an on-line [NOVAP](#) or [RHoliday](#) form, respectively, at least a week ahead of time and decide with the instructor how to make up the work.
- Those with a personal emergency or bereavement should inform their academic dean and the instructors. Please contact the instructor as soon as possible after returning to schedule make-up work.

Testing center

This class will use the [Testing Center](#) to provide testing accommodations and temporary test-taking supports to undergraduates registered with and approved by the Student Disability Access Office (SDAO) and/or Academic Resource Center (ARC). The Testing Center operates by appointment only and appointments must be made at least 7 consecutive days in advance, but please schedule your appointments as far in advance as possible. You will not be able to make an appointment until you have submitted a Semester Request with the SDAO or completed screening at the ARC and your accommodations or supports have been approved. If you have not already done so, promptly submit a Semester Request to the SDAO or schedule your screening with the ARC in order to make your appointment in time. For instructions on how to register with SDAO, visit their website at <https://access.duke.edu/requests>. For instructions on how to schedule a screening with the ARC, visit their website at <https://arc.duke.edu/ld-adhd/overview>. For instructions on how to make an appointment at the Testing Center, visit their website at <https://testingcenter.duke.edu>.

Academic integrity

Academic dishonesty on tests, plagiarism on homework and projects, copying homework, lying about an illness or absence and other forms of academic dishonesty are a breach of trust with classmates and faculty and will not be tolerated. They also violate Duke's [Community Standard](#) and may be referred to the Office of Student Conduct as described here in the [Academic Integrity Council](#). Additionally, there may be penalties applied to the assignment(s) in question and/or to the final course grade.